Homework 06: Network duality and integer production

ISyE 524 — Fall 2026
Due: Monday, October 26, 2026, at 11:59 p.m. (Madison time).

Complete all three problems. Problems 1–2 revisit the networks from HW05, now focusing on duality, complementary slackness, and optimality certificates. All data needed for those problems are repeated here. Problem 3 uses batch production, integer variables, LP relaxation, and rounding from the opening integer-programming material.

Write your mathematical formulation before implementing it in Julia/JuMP. Use HiGHS, and check solver status and solution availability before reading values. Problems 1–2 are continuous LPs; only Problem 3(a) requires integer variables. Keep each written answer and its reasoning together.

Submit **one PDF to Gradescope** and mark which pages belong to each problem. **You will not receive full marks unless you mark the pages.**

Getting started and written answers

Follow Assignments and PDF Submission. In VS Code, use **Tasks: Run Task** to run **ISyE 524: Update course repository**, then **ISyE 524: Start an assignment from its template**, entering `hw06`. Open `student-work/hw06/hw06.ipynb`. An existing working copy will not be overwritten.

For **every written part**, type LaTeX mathematics in Markdown or insert a clear photograph/scan of handwritten work. Include the answer and reasoning together and label the subparts; no duplicate transcription is required. Julia/JuMP implementations belong in code cells.

Use `$x_1 \geq 0$` for inline mathematics and `$$x_1+x_2\leq5$$` for display mathematics. Run the Markdown cell to render it.

For handwritten work, edit a Markdown cell in VS Code, drag in a cropped PNG/JPEG, and choose **Insert Image as Attachment**. Alternatively, put the image beside your working notebook under `student-work/hw06/images/` and insert `![My answer](images/problem1.png)`. Keep external images with the notebook. See Images and handwritten work.

Back up `student-work/hw06/` separately; Git does not back up your answers or images. After completing the notebook, export it with the course PDF task, inspect `submissions/hw06.pdf` for readable mathematics, tables, outputs, and images, and submit that PDF to Gradescope.

Computational setup

Use the course Julia environment. The cells below supply the network data; complete the model and reporting cells. They run without unfinished answers. For Problems 1–2, implement the **dual LPs you derive**, rather than reading dual values from a primal solve. No graph-search algorithm is required.

```
using JuMP, HiGHS, Printf
import MathOptInterface as MOI
using DataFrames
```

1: Taste of Madison — the shortest-path dual

You want to drive from **Short Stack (S)** to **Alchemy (A)**. Intermediate stops are optional. The other restaurants are **Great Dane (G)**, **Old Fashioned (O)**, **Red Sushi (R)**, **Ian's at Garver (I)**, and **Weary Traveler (W)**. Only the directed arcs below are available. Travel times c_{ij} are in minutes.

Arc	Time	Arc	Time	Arc	Time
S → G	4	G → I	7	R → W	5
S → O	6	O → R	2	I → W	1
S → R	5	O → I	5.5	I → A	6
G → O	1	O → W	4	W → A	8

Let $\mathcal{N} = \{S, G, O, R, I, W, A\}$ and let $\mathcal{E}$ be these twelve arcs. Define $b_S = 1$, $b_A = -1$, and $b_i = 0$ otherwise. With continuous flow variables $x_{ij} \geq 0$, the HW05 shortest-path LP is

$$\begin{aligned} \min_{x} \quad & \sum_{(i,j)\in\mathcal{E}} c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{j:(i,j)\in\mathcal{E}} x_{ij} - \sum_{j:(j,i)\in\mathcal{E}} x_{ji} = b_i \quad i \in \mathcal{N}, \\ & x_{ij} \geq 0 \qquad\qquad\qquad\qquad (i,j) \in \mathcal{E}. \end{aligned}$$

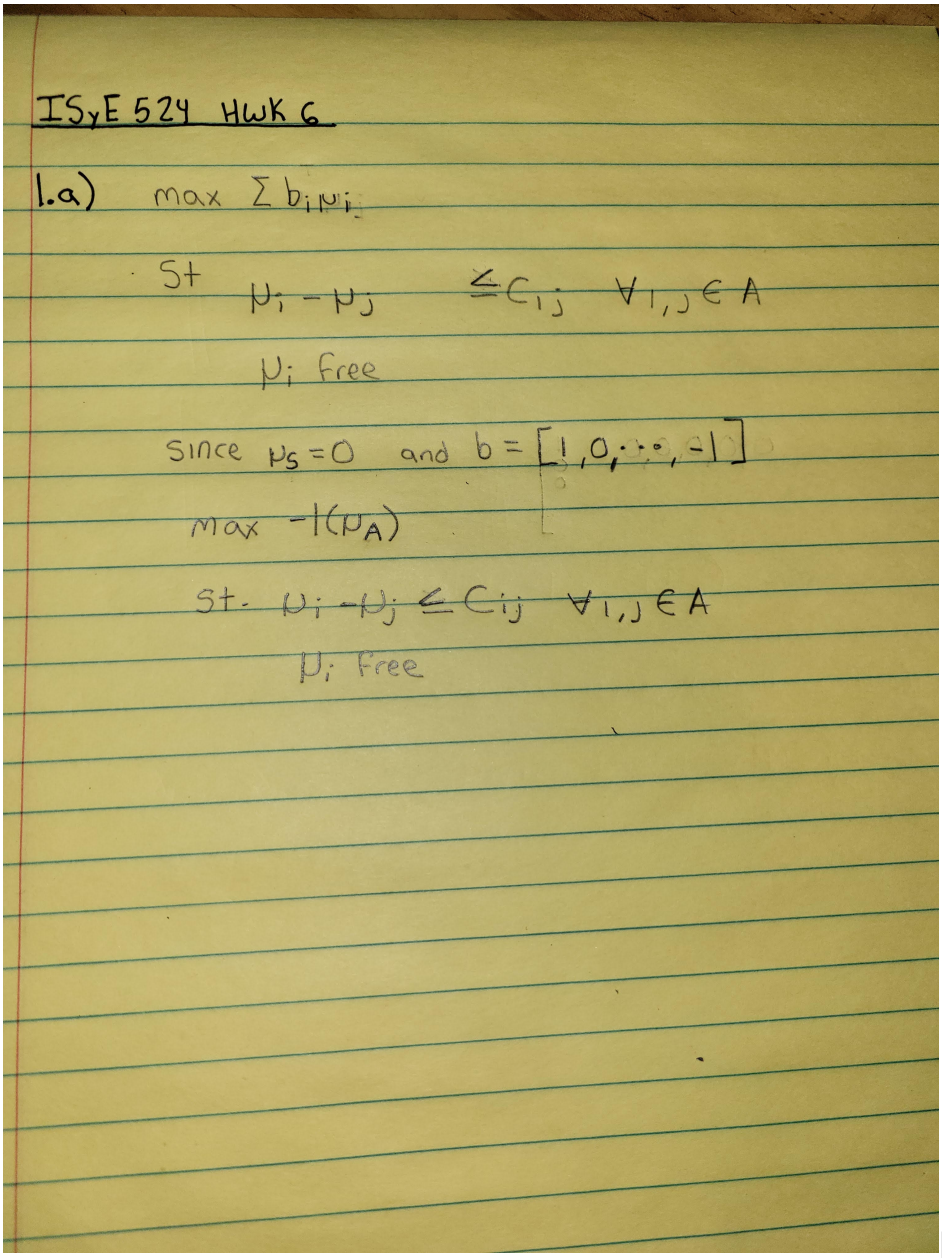
There are no finite arc-capacity constraints in this model.

1(a): Derive the dual

Use a **free node potential** μ_i for each flow-balance equality. Derive the dual LP in indexed form, including its objective, arc constraints, and variable domains. Explain why adding the normalization $\mu_S = 0$ does not change its optimal value.

Your answer and reasoning:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]



Adding the normalization $\mu_S = 0$ does not change its optimal value because it just just gives the problem a refrence value. The flows will remain the same if you set it to another number the optimal value will remain the same because it just cancels out in the objective function when you expand it out. This is because μ_A will be greater by the value of μ_S and because the objective is $\mu_S - \mu_A$ the value irrelevent to the final optimal soloution.

1(b): Solve the dual

Implement the dual from 1(a) with $\mu_S = 0$ and solve it. Report the objective value and a table of node potentials. Your code should use `mu[i]` for μ_i. Different optimal potential vectors are acceptable.

```
nodes_path = [:S, :G, :O, :R, :I, :W, :A]
arcs_path = [(:S,:G), (:S,:O), (:S,:R), (:G,:O), (:G,:I),
             (:O,:R), (:O,:I), (:O,:W), (:R,:W), (:I,:W),
             (:I,:A), (:W,:A)]
c_path = Dict(zip(arcs_path, [4.0,6.0,5.0,1.0,7.0,2.0,
                              5.5,4.0,5.0,1.0,6.0,8.0]))
b_path = Dict(i => (i == :S ? 1.0 : i == :A ? -1.0 : 0.0)
              for i in nodes_path);
```

```
# Build and solve the dual using free variables mu[nodes_path].
# Normalize mu[:S] to zero; report the objective and node potentials.

MCNF = Model(HiGHS.Optimizer)

@variable(MCNF, mu[i in nodes_path])

@objective(MCNF, Max, -mu[:A])

@constraint(MCNF, flowbalance[(i,j) in arcs_path],
    mu[i]-mu[j] <= c_path[i,j])

@constraint(MCNF, mu[:S] == 0)

optimize!(MCNF)
status = termination_status(MCNF)
println("Termination status: ", status)
status == MOI.OPTIMAL || error("HiGHS stopped with status $(status).")
is_solved_and_feasible(MCNF) || error("No feasible optimal solution is available

shortest_time = objective_value(MCNF)

println("Shortest time is: ", shortest_time, " Minuites")

node_potentials = DataFrame(
    flows = -(value.(mu[(i)] for i in nodes_path)),
    node= collect(nodes_path),
)

display(node_potentials)
```

```
Running HiGHS 1.15.1 (git hash: 04024d701f): Copyright (c) 2026 under MIT licenc
Includes third-party software components, see THIRD_PARTY_NOTICES.md for full de
Using BLAS: blastrampoline
LP has 13 rows; 7 cols; 25 nonzeros
Coefficient ranges:
  Matrix  [1e+00, 1e+00]
  Cost    [1e+00, 1e+00]
```

```
  Bound   [0e+00, 0e+00]
  RHS     [1e+00, 8e+00]
Presolving model
0 rows, 0 cols, 0 nonzeros 0s
0 rows, 0 cols, 0 nonzeros 0s
Presolve reductions: rows 0(-13); columns 0(-7); nonzeros 0(-25) - Reduced to em
Performed postsolve
Solving the original LP from the solution after postsolve

Model status        : Optimal
Objective value     :  1.6500000000e+01
P-D objective error :  0.0000000000e+00
HiGHS run time      :          0.00
Termination status: OPTIMAL
Shortest time is: 16.5 Minuites
```

	flows	node
	Float64	Symbol
1	0.0	S
2	4.0	G
3	5.0	O
4	5.0	R
5	10.5	I
6	9.0	W
7	16.5	A

Your reported results:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

The shortest time for this network is 16.5 minuites.

Node Potential	Value (minuites)
S	0
G	4
O	5
R	5
I	10.5
W	9
A	16.5

1(c): Interpret the potentials and certify a route

For any feasible potential vector normalized by $\mu_S = 0$, add the arc inequalities along an arbitrary directed path from S to a node i. Use the resulting inequality to interpret $-\mu_i$. Must it equal the shortest S–i distance at **every** node in an optimal dual solution? Explain briefly.

Use the route **S–G–O–I–A** to give an optimality certificate: compute its travel time, check that its arcs are tight in your optimal dual solution, and explain why the route and potential vector together prove optimality. State the complementary-slackness condition being used.

Your answer and reasoning:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

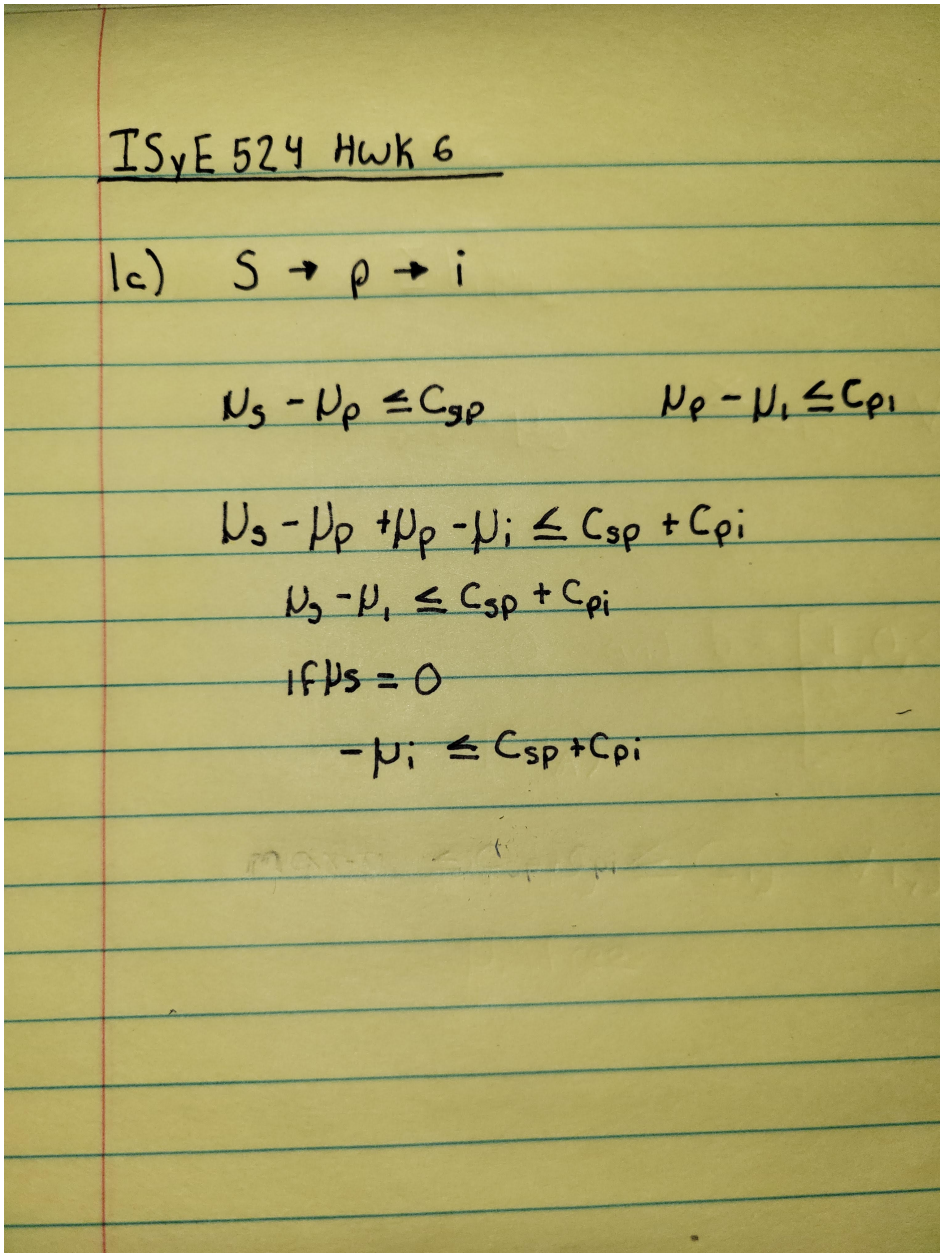


Figure 1: image.png

$-\mu_i$ must equal the shortest S–i distance at every node in an optimal dual solution becasue it is bounded by the sums of the arc distances on the path and complimentary slackness means that the equality will become tight on the optimal path and $-\mu_i = C_{sp} + C_{pi}$

```
S_G=-(node_potentials.flows[1]-node_potentials.flows[2])
G_O=-(node_potentials.flows[2]-node_potentials.flows[3])
O_I=-(node_potentials.flows[3]-node_potentials.flows[5])
I_A=-(node_potentials.flows[5]-node_potentials.flows[7])

println("S to G: ",S_G," G to O: ",G_O," O to I: ",O_I," I to A: ",I_A)

if S_G - c_path[:S,:G] ==0
    println("S to G constraint is tight")
end
if G_O - c_path[:G,:O] ==0
    println("G to O constraint is tight")
end
if O_I - c_path[:O,:I] ==0
    println("O to I constraint is tight")
end
if I_A - c_path[:I,:A] ==0
    println("I to A constraint is tight")
end
```

```
S to G: 4.0 G to O: 1.0 O to I: 5.5 I to A: 6.0
S to G constraint is tight
G to O constraint is tight
O to I constraint is tight
I to A constraint is tight
```

Becasue every constraint in S-G-O-I-A is tight it means that the all other primal variables are 0 because we only send 1 through the network and every node with a tight constraint will have values $ >0$. This is because the 1 flow sent must flow through the tight constraints and 0 must flow through the non tight constraints. The complimentary slackness rule used in this case is that if the constraint isnt tight it must have 0 flow.

2: Maximum flow — a minimum-cut certificate

Revisit the six-node network from HW05. Each arc label is its capacity in units of flow per hour. Intermediate nodes neither create nor consume flow.

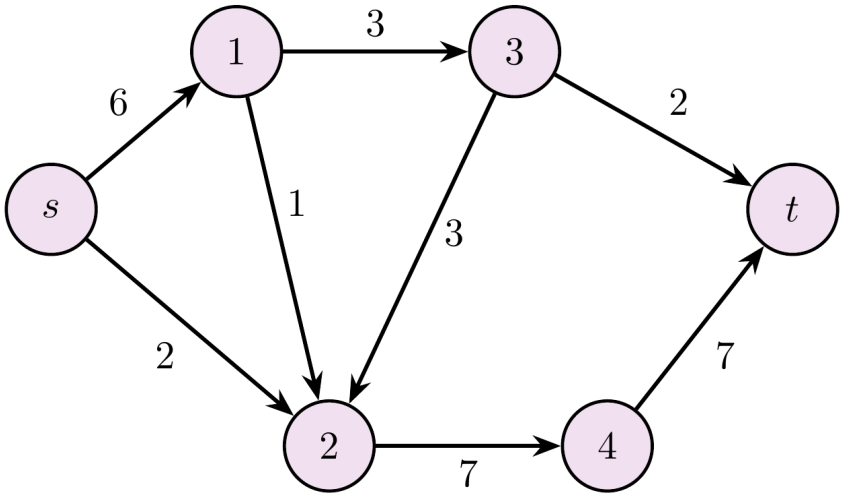


Figure 2: The six-node network, with capacities on its directed arcs.

Let $\mathcal{N} = \{s, 1, 2, 3, 4, t\}$ and let $\mathcal{E}$ contain only the eight physical arcs shown in the figure. Continuous variables $x_{ij} \geq 0$ denote arc flows, and $v \geq 0$ denotes the flow on an additional feedback arc (t, s). For deriving the dual, write maximum flow in the following equivalent **maximization** form:

$$\begin{aligned} \max_{x,v} \quad & v \\ \text{s.t.} \quad & \sum_{j:(i,j)\in\mathcal{E}} x_{ij} - \sum_{j:(j,i)\in\mathcal{E}} x_{ji} + \alpha_i v = 0 \quad i \in \mathcal{N}, \\ & x_{ij} \leq u_{ij} \qquad\qquad (i,j) \in \mathcal{E}, \\ & x_{ij} \geq 0 \qquad\qquad (i,j) \in \mathcal{E}, \qquad v \geq 0, \end{aligned}$$

where $\alpha_s = -1$, $\alpha_t = 1$, and $\alpha_i = 0$ at intermediate nodes. The feedback arc has no finite capacity. This maximizes the same throughput as HW05's minimization of $-v$.

2(a): Derive the dual

Use **free node potentials** π_i for the balance equalities and **nonnegative multipliers** λ_{ij} for the physical-arc capacity constraints. Write the dual LP in indexed form. Include the constraint arising from $v \geq 0$ and all variable domains. You may normalize $\pi_s = 0$.

Your answer and reasoning:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

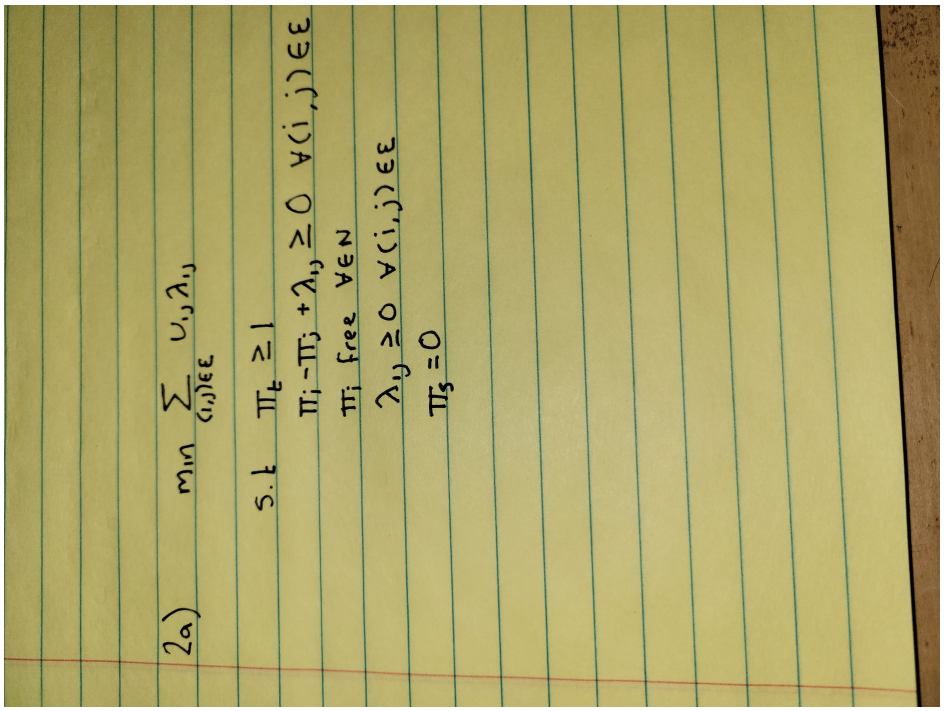


Figure 3: 1000006494.jpg

2(b): Construct a complementary dual solution

The following physical-arc flow is supplied:

Arc	Capacity u_{ij}	Flow x_{ij}
s $\to$ 1	6	4
s $\to$ 2	2	2
1 $\to$ 2	1	1
1 $\to$ 3	3	3
3 $\to$ 2	3	1
2 $\to$ 4	7	4
3 $\to$ t	2	2
4 $\to$ t	7	4

Check its capacities and node balances, and determine the feedback flow v. Starting with $\pi_s = 0$, **use complementary slackness by hand** to construct a dual solution complementary to this flow. Show which multipliers must be zero and which dual constraints must be tight. Check dual feasibility and compute the dual objective.

Then implement and solve your dual LP in Julia/JuMP to check the optimal value. Use `pi[i]` and `lambda[a]` for π_i and λ_{ij}. Report the value and one optimal dual vector; solver output does not replace your complementary-slackness argument.

Your answer and reasoning:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

Arc	Capacity u_{ij}	Flow x_{ij}	Slack/Tight
s $\rightarrow$ 1	6	4	Slack
s $\rightarrow$ 2	2	2	Tight
1 $\rightarrow$ 2	1	1	Tight
1 $\rightarrow$ 3	3	3	Tight
3 $\rightarrow$ 2	3	1	Slack
2 $\rightarrow$ 4	7	4	Slack
3 $\rightarrow$ t	2	2	Tight
4 $\rightarrow$ t	7	4	Slack

-

Node	Inflow	Outflow
1	4	4
2	4	4
3	3	3
4	4	4

$v = x_{s1} + x_{s2}$

$v= 4+2 = 6 $

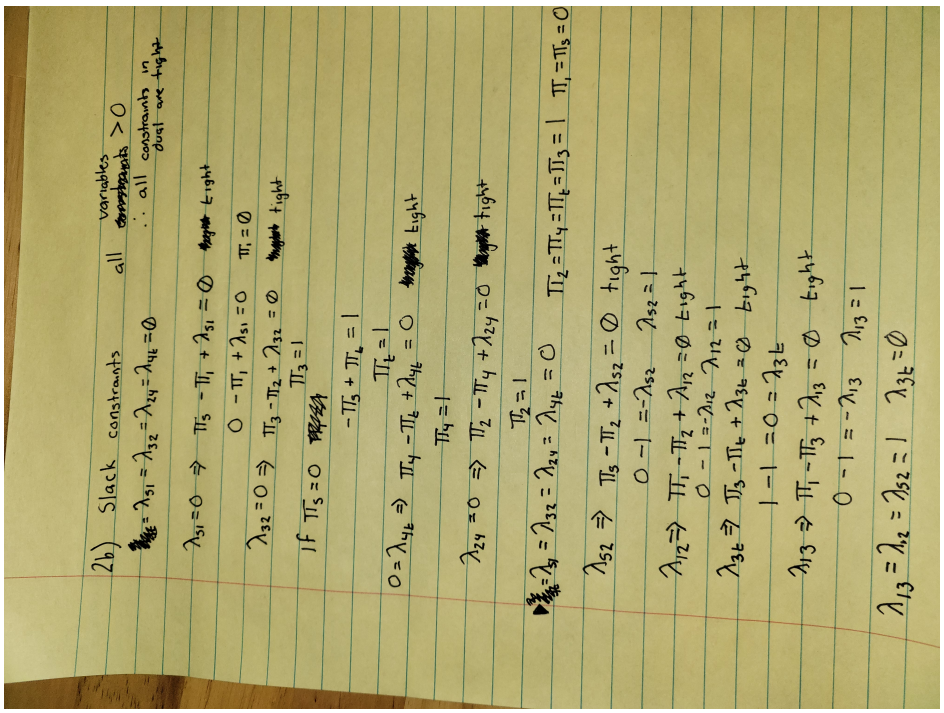


Figure 4: 1000006495.jpg

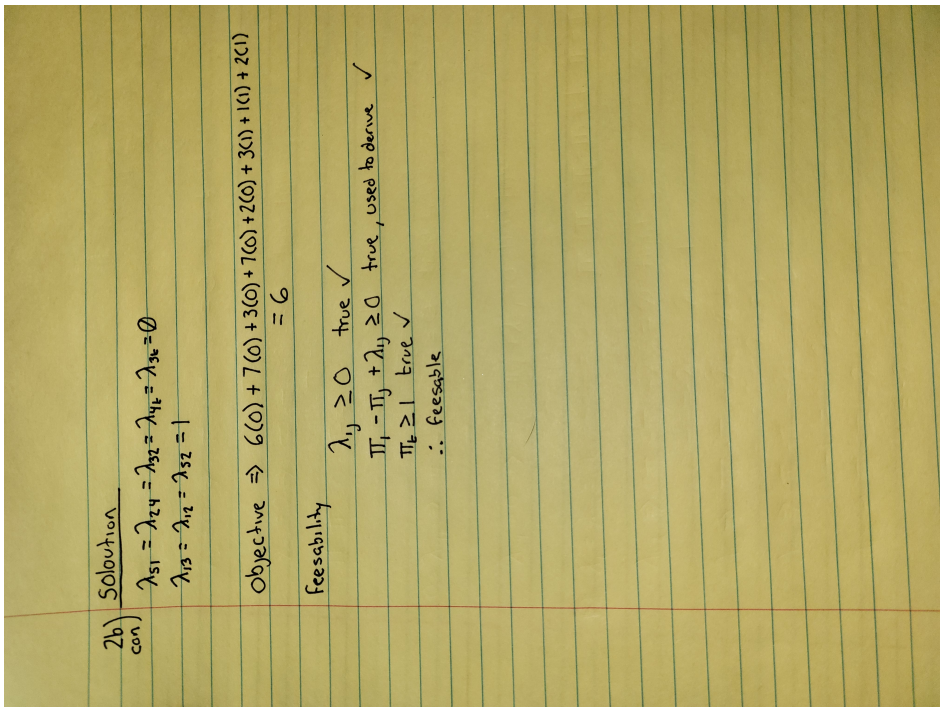


Figure 5: 1000006496.jpg

```
# Strings "1", ..., "4" label the intermediate nodes in the figure.
nodes_flow = ["s", "1", "2", "3", "4", "t"]
arcs_flow = [("s","1"), ("s","2"), ("1","2"), ("1","3"),
             ("3","2"), ("2","4"), ("3","t"), ("4","t")]
u_flow = Dict(zip(arcs_flow, [6.0,2.0,1.0,3.0,3.0,7.0,2.0,7.0]))
x_given = Dict(zip(arcs_flow, [4.0,2.0,1.0,3.0,1.0,4.0,2.0,4.0]));
```

```
# Build and solve the dual with free pi and nonnegative lambda.
# Normalize pi["s"] to zero; report the objective and optimal vectors.


MCNF2 = Model(HiGHS.Optimizer)

@variable(MCNF2, lamda[(i,j) in arcs_flow] >= 0)

@variable(MCNF2, pi[i in nodes_flow])

@objective(MCNF2, Min, sum(u_flow[(i,j)]*lamda[(i,j)] for (i,j) in arcs_flow))

@constraint(MCNF2, flowbalance[(i,j) in arcs_flow],
    pi[i]-pi[j]+lamda[(i,j)]>=0)

@constraint(MCNF2, pi[:"t"] >= 1)

@constraint(MCNF2, pi[:"s"] == 0)
```

```
optimize!(MCNF2)
status = termination_status(MCNF2)
println("Termination status: ", status)
status == MOI.OPTIMAL || error("HiGHS stopped with status $(status).")
is_solved_and_feasible(MCNF2) || error("No feasible optimal solution is availabl

maximum_flow = objective_value(MCNF2)

println("Maximum Flow is: ", maximum_flow, " Products per Hr")

node_potentials = DataFrame(
    flows = (value.(lamda[(i,j)] for (i,j) in arcs_flow)),
    arcs= collect(arcs_flow),
)

display(node_potentials)
```

```
Running HiGHS 1.15.1 (git hash: 04024d701f): Copyright (c) 2026 under MIT licenc
Includes third-party software components, see THIRD_PARTY_NOTICES.md for full de
Using BLAS: blastrampoline
LP has 10 rows; 14 cols; 26 nonzeros
Coefficient ranges:
  Matrix  [1e+00, 1e+00]
  Cost    [1e+00, 7e+00]
  Bound   [0e+00, 0e+00]
  RHS     [1e+00, 1e+00]
Presolving model
8 rows, 12 cols, 20 nonzeros 0s
4 rows, 8 cols, 15 nonzeros 0s
2 rows, 3 cols, 4 nonzeros 0s
Presolve reductions: rows 2(-8); columns 3(-11); nonzeros 4(-22)
Solving the presolved LP
Using dual simplex solver
  Iteration        Objective     Infeasibilities num(sum)
          0     3.0000000000e+00 Pr: 2(2) 0.0s
          1     6.0000000000e+00 Pr: 0(0) 0.0s

Performed postsolve
Solving the original LP from the solution after postsolve

Model status        : Optimal
Simplex   iterations: 1
Objective value     :  6.0000000000e+00
P-D objective error :  0.0000000000e+00
HiGHS run time      :          0.00
```

```
Termination status: OPTIMAL
Maximum Flow is: 6.0 Products per Hr
```

	flows	arcs
	Float64	Tuple…
1	0.0	(”s”, ”1”)
2	1.0	(”s”, ”2”)
3	1.0	(”1”, ”2”)
4	1.0	(”1”, ”3”)
5	0.0	(”3”, ”2”)
6	0.0	(”2”, ”4”)
7	0.0	(”3”, ”t”)
8	0.0	(”4”, ”t”)

Maximum flow is 6 products per hour with values of 1 on arc S-2, 1-2 and 1-3 with 0 on the rest.

2(c): Identify the cut and explain the certificate

Use your constructed potentials from 2(b) to specify a cut $(S, \overline{S})$, where $s \in S$ and $t \in \overline{S}$. List the **physical arcs directed from S to $\overline{S}$** and sum their capacities.

Explain why the capacity of any such cut is an upper bound on every feasible s–t flow. Use your cut and the supplied feasible flow to conclude that the flow is maximum and the cut is minimum. Relate the cut arcs to your capacity multipliers λ_{ij}.

Your answer and reasoning:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

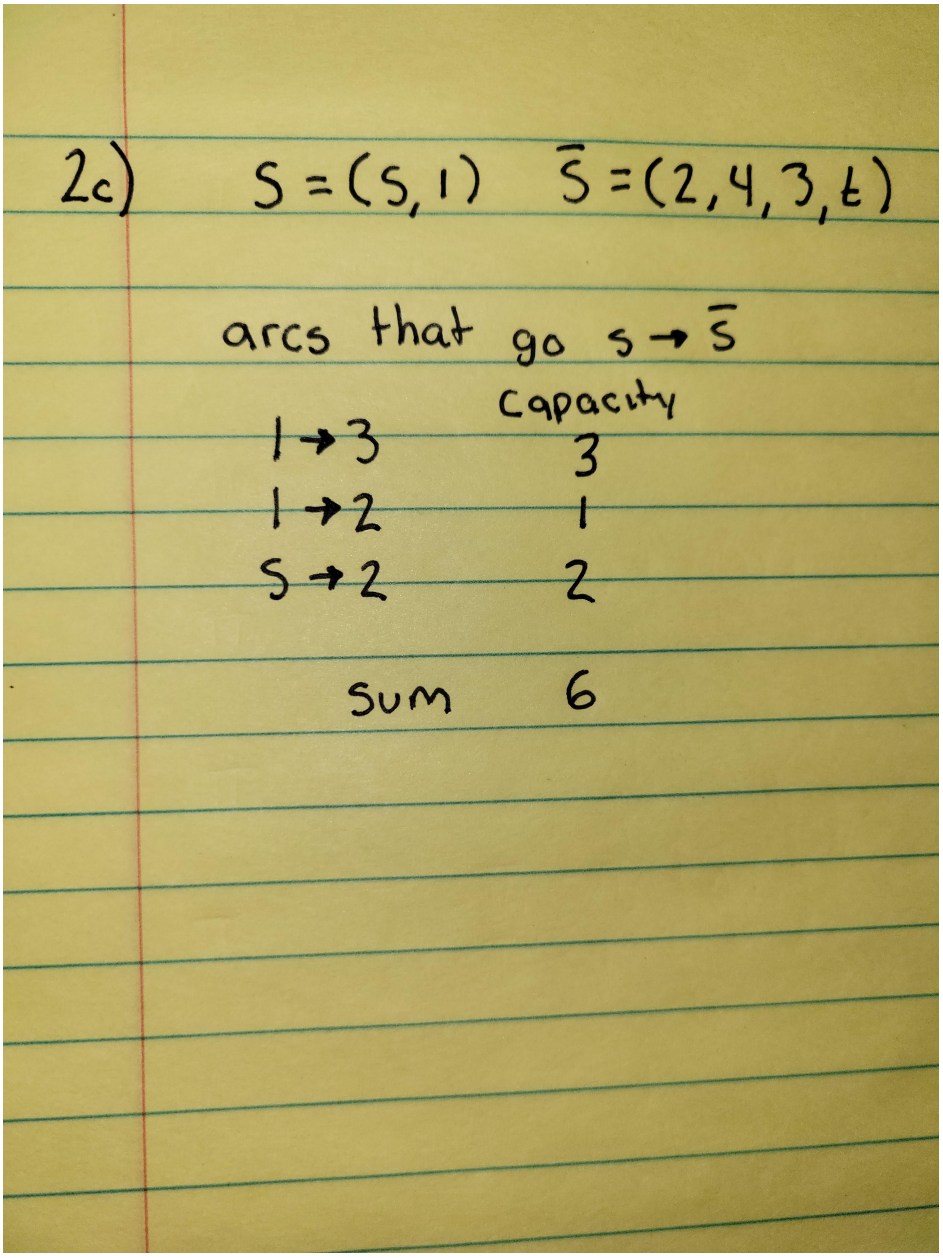


Figure 6: image.png

This has to be an upper bound of the flow becasue any flow must pass through a cut from S to $\overline{S}$. This means the max flow must have an upper bound of every cut. Becasue this cut seperates $\pi_i = 1$ and $\pi_i = 0$ it cuts through the arcs with $\lambda_{ij} = 1$ which means their capacities summed gives the optimal soloution or the minimum of the cuts values.

3: Campus Awards — production in whole batches

A small workshop makes football trophies and soccer trophies. Production must occur in complete batches, using the resources below.

Product	Trophies per batch	Labor per batch (hours)	Material per batch (kg)	Profit per batch ($)
Football trophies	10	3	2	40
Soccer trophies	20	2	4	50

For the coming week, **19 labor hours** and **23 kg of material** are available. All completed trophies can be sold. Profit in the table is net of variable production costs; there are no setup costs. No fractional batches are permitted, and no minimum production is required.

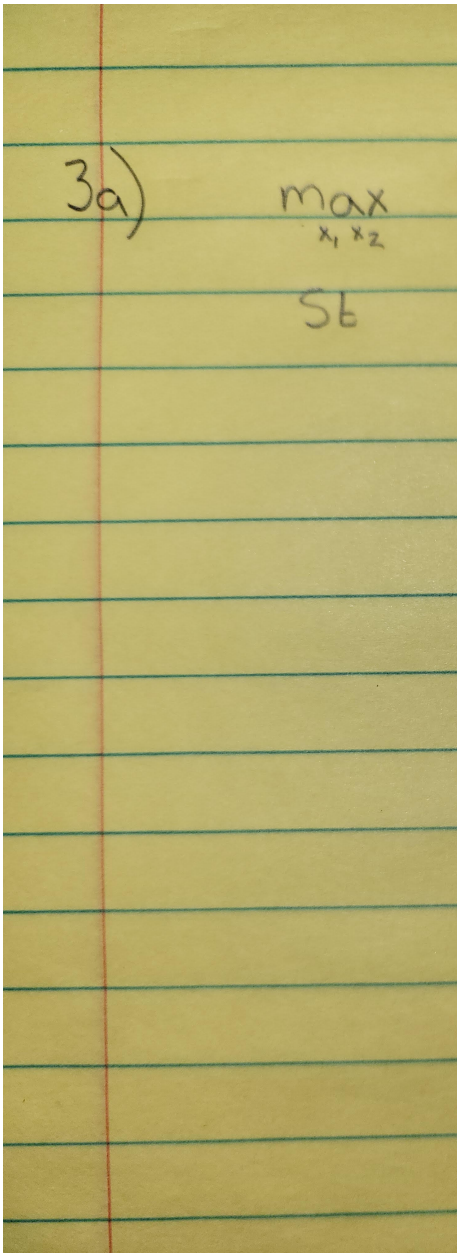
Use these scalar decision variables:

- $x_1 \in \mathbb{Z}_{\geq 0}$: the number of complete football-trophy batches.
- $x_2 \in \mathbb{Z}_{\geq 0}$: the number of complete soccer-trophy batches.

3(a): Formulate and solve the integer program

Write the profit-maximizing IP, including both resource constraints and variable domains. Implement it with scalar JuMP variables `x1` and `x2`. Report the optimal batch counts, numbers of trophies, total profit, and labor and material used.

Your formulation and reasoning:



[Type your answer here, or insert a clear photograph/scan of handwritten work.]

```
# Formulate and solve the IP with scalar integer variables x1 and x2.
# Report batches, trophies, profit, and resource use.

Integer = Model(HiGHS.Optimizer)

@variable(Integer, x1>=0, Int)

@variable(Integer, x2>=0, Int)

@objective(Integer, Max, 40*x1+50*x2)

@constraint(Integer, 3*x1+2*x2 <= 19)
```

```julia
@constraint(Integer, 2*x1+4*x2 <= 23)

optimize!(Integer)
status = termination_status(Integer)
println("Termination status: ", status)
status == MOI.OPTIMAL || error("HiGHS stopped with status $(status).")
is_solved_and_feasible(Integer) || error("No feasible optimal solution is availa

maximum_profit = objective_value(Integer)
football_batches = value.(x1)
soccer_batches = value.(x2)

println("The maximum profit is: ", maximum_profit, " dollars")
println("Football Batches: ", football_batches)
println("Soccer Batches: ", soccer_batches)
println("Football Number of trophies: ", football_batches*10)
println("Soccer Number of trophies: ", soccer_batches*20)
println("Labor Used: ", football_batches*3 + 2*soccer_batches, " hr")
println("Material Used: ", football_batches*2 + 4*soccer_batches, " kg")
```

```
Running HiGHS 1.15.1 (git hash: 04024d701f): Copyright (c) 2026 under MIT licenc
Includes third-party software components, see THIRD_PARTY_NOTICES.md for full de
Using BLAS: blastrampoline
MIP has 2 rows; 2 cols; 4 nonzeros; 2 integer variables (0 binary)
Coefficient ranges:
  Matrix  [2e+00, 4e+00]
  Cost    [4e+01, 5e+01]
  Bound   [0e+00, 0e+00]
  RHS     [2e+01, 2e+01]
Presolving model
2 rows, 2 cols, 4 nonzeros 0s
2 rows, 2 cols, 4 nonzeros 0s
Presolve reductions: rows 2(-0); columns 2(-0); nonzeros 4(-0)
Objective function is integral with scale 0.1

Solving MIP model with:
   2 rows
   2 cols (0 binary, 2 integer, 0 implied int., 0 continuous, 0 domain fixed)
   4 nonzeros
   Thread count 2 (of 4 threads). Using 1 max workers. Parallel search off

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristi
     I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Ra
     S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y =>
     Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper;
```

```
        Nodes      |    B&B Tree     |            Objective Bounds
Src  Proc. InQueue |  Leaves   Expl. | BestBound       BestSol              Gap

 z       0       0         0   0.00%   inf             40                 Large
 J       0       0         0   0.00%   inf             290                Large
 S       0       0         0   0.00%   400             310              29.03%
         0       0         0   0.00%   335             310               8.06%
 T       0       0         0   0.00%   335             320               4.69%
         1       0         1 100.00%   320             320               0.00%

Solving report
  Status            Optimal
  Primal bound      320
  Dual bound        320
  Gap               0% (tolerance: 0.01%)
  P-D integral      0.000110819836347
  Solution status   feasible
                    320 (objective)
                    0 (bound viol.)
                    0 (int. viol.)
                    0 (row viol.)
  Timing            0.01
                    0.00 (Presolve)
                    0.01 (Solve)
                    0.00 (Postsolve)
  Max sub-MIP depth 0
  Nodes             1
  Repair LPs        0
  LP iterations     3
                    0 (strong br.)
                    1 (separation)
                    0 (heuristics)
Termination status: OPTIMAL
The maximum profit is: 320.0 dollars
Football Batches: 3.0
Soccer Batches: 4.0
Football Number of trophies: 30.0
Soccer Number of trophies: 80.0
Labor Used: 17.0 hr
Material Used: 22.0 kg
```

The optimal batch counts are 3 football and 4 soccer batches. The optimal numbers of football trophies are 30 and the optimal number of soccer trophies is 80. Total profit is 320 dollars using 17hrs labor and 22kg material.

3(b): Solve the LP relaxation

Replace the integer domains by $x_1, x_2 \geq 0$ continuous and solve the resulting LP. Report its batch counts and profit. Explain why this LP value is an **upper bound** on the IP optimum and why its production plan need not be implementable. Compute the difference between the LP and IP objective values in dollars.

```
# Solve the same model with continuous scalar variables x1 and x2.
# Report the relaxed plan, objective, and dollar difference from the IP.
Relax = Model(HiGHS.Optimizer)

@variable(Relax, x1>=0)

@variable(Relax, x2>=0)

@objective(Relax, Max, 40*x1+50*x2)

@constraint(Relax, 3*x1+2*x2 <= 19)

@constraint(Relax, 2*x1+4*x2 <= 23)

optimize!(Relax)
status = termination_status(Relax)
println("Termination status: ", status)
status == MOI.OPTIMAL || error("HiGHS stopped with status $(status).")
is_solved_and_feasible(Relax) || error("No feasible optimal solution is availabl

maximum_profit_relax = objective_value(Relax)
football_batches_relax = value.(x1)
soccer_batches_relax = value.(x2)

println("The maximum profit is: ", maximum_profit_relax, " dollars")
println("Football Batches: ", football_batches_relax)
println("Soccer Batches: ", soccer_batches_relax)
println("Football Number of trophies: ",
 football_batches_relax*10)
println("Soccer Number of trophies: ",
 soccer_batches_relax*20)
println("Labor Used: ",
 football_batches_relax*3 + 2*soccer_batches_relax, " hr")
println("Material Used: ",
 football_batches_relax*2 + 4*soccer_batches_relax, " kg")
```

```
Running HiGHS 1.15.1 (git hash: 04024d701f): Copyright (c) 2026 under MIT licenc
Includes third-party software components, see THIRD_PARTY_NOTICES.md for full de
Using BLAS: blastrampoline
LP has 2 rows; 2 cols; 4 nonzeros
```

```
Coefficient ranges:
  Matrix  [2e+00, 4e+00]
  Cost    [4e+01, 5e+01]
  Bound   [0e+00, 0e+00]
  RHS     [2e+01, 2e+01]
Presolving model
2 rows, 2 cols, 4 nonzeros 0s
2 rows, 2 cols, 4 nonzeros 0s
Presolve reductions: rows 2(-0); columns 2(-0); nonzeros 4(-0) - Not reduced
Problem not reduced by presolve: solving the LP
Using dual simplex solver
  Iteration        Objective     Infeasibilities num(sum)
          0    -8.9999930051e+01 Ph1: 2(11); Du: 2(89.9999) 0.0s
          2     3.4375000000e+02 Pr: 0(0) 0.0s

Model status        : Optimal
Simplex   iterations: 2
Objective value     :  3.4375000000e+02
P-D objective error :  0.0000000000e+00
HiGHS run time      :          0.00
Termination status: OPTIMAL
The maximum profit is: 343.75 dollars
Football Batches: 3.75
Soccer Batches: 3.875
Football Number of trophies: 37.5
Soccer Number of trophies: 77.5
Labor Used: 19.0 hr
Material Used: 23.0 kg
```

Your results and explanation:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

After relaxing the constraints to allow for non intiger batches the maximum profit becomes 343.75 dollars with 3.75 football batches and 3.875 soccer batches. This is more profit than the intiger case because it allow for making only partial batches of each trophy and the result is making 7.5 extra football trophies and 2.5 less soccer trophies but it fully utilizes the available labor and material unlike the intiger soloution which still had capacitiy left over but not enough to make another batch of either trophy. This maximum profit becomes an upper bound for the IP because removing constriants can only increace or not change the final soloution in maximization problem it cannot make it smaller.

3(c): Does rounding repair the relaxed solution?

Starting from the relaxed batch counts in 3(b), consider two plans:

1. Round **each** batch count to its nearest integer.
2. Round **each** batch count down to the next lower integer.

For each plan, report the batch counts and resource requirements, and check feasibility. For each feasible plan, report its profit and compare it with the IP optimum. What does this example show about using rounding as a replacement for solving the IP? You may calculate these quantities by hand or use the optional code cell below.

```
# Optional: compare nearest-integer and rounded-down plans.
```

Your answer and reasoning:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

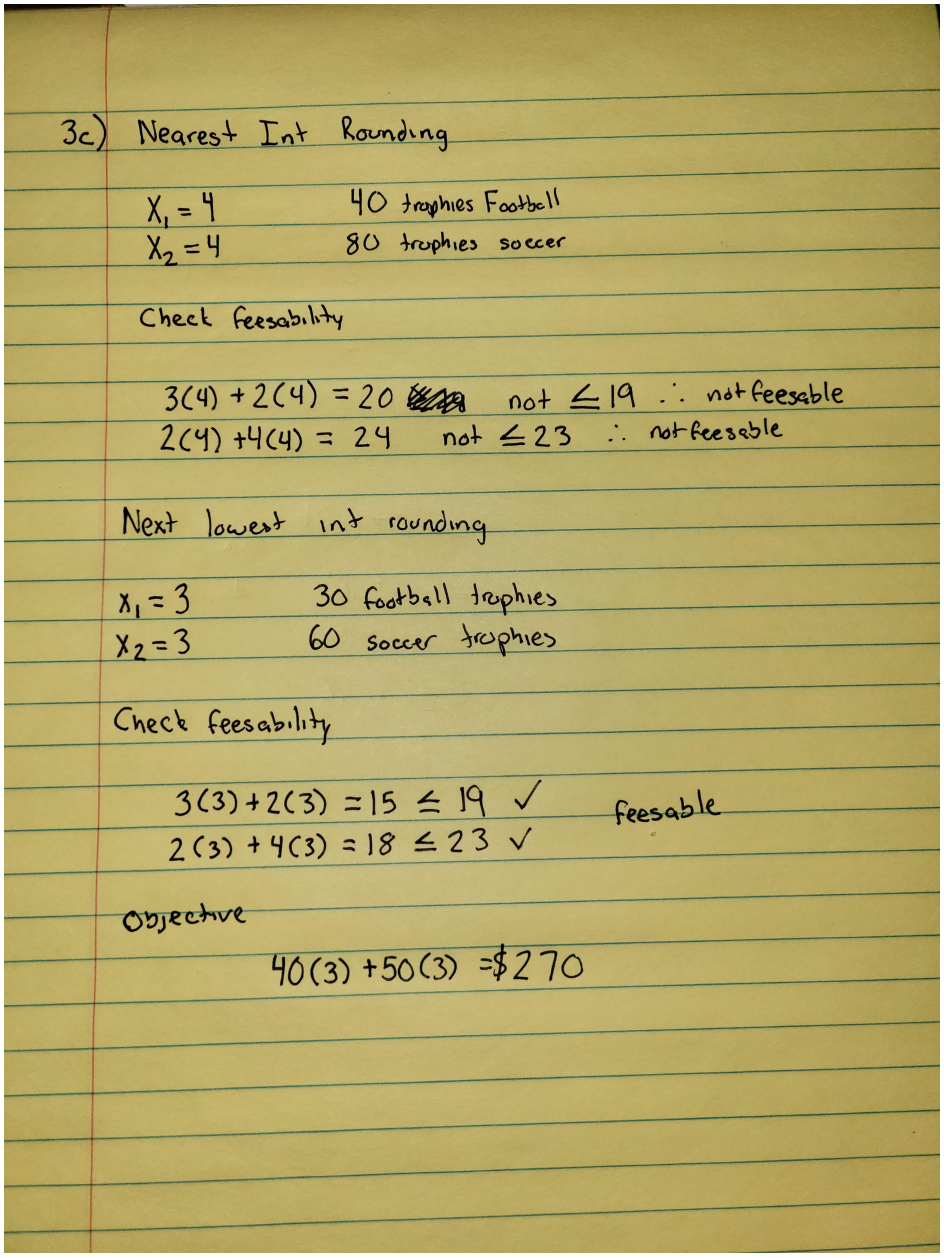


Figure 7: image.png

For each plan, report the batch counts and resource requirements, and check feasibility. For each feasible plan, report its profit and compare it with the IP optimum. What does this example show about using rounding as a replacement for solving the IP? You may calculate these quantities by hand or use the optional code cell below.

For the Nearest integer rounding there would be 4 batches of each football and soccer trophies, this would require 20 hrs labor and 24kg material which makes it not feasable. For the next lowest integer rounding there would be 3 batches of each football and soccer trophies. This results in needing 15 hrs labor and 18kg material and a profit of 270 dollars. This shows that you can use rounding to find close soloutions for your optimal integer soloution but cannot gaurentee that it is

the optimal since in this case the optimal was 3 batches football and 4 soccer which would require rounding up on soccer but down on football.

Collaboration and LLM-use statement — required

Complete this statement as the **very last part of your homework**. Be specific enough that a reader can understand what work you did yourself and how others or LLMs helped you learn.

- **People:** Give the name of every person you collaborated with, identify the problem/subpart you discussed, and explain the nature of the collaboration. If you worked alone, explicitly state that you did not collaborate with anyone.
- **LLMs/tools:** Name every LLM or AI assistant you used and identify the model/version as precisely as available. If the interface did not show a model/version, say so. If you used none, explicitly state that you did not use an LLM or AI assistant.
- **Specific assistance and learning:** For each tool, identify the problem/subpart, describe what you asked it to help with, what suggestions you used or rejected, and how the interaction helped you understand the mathematics or implementation. Explain how you checked the suggestions. "I used ChatGPT for help" is not sufficiently specific.

Disclosure does not replace the requirement that your submitted formulation, implementation, and explanation be your own work and that you understand them.

People and collaboration:

[Names, problem/subparts, and what you discussed; or explicitly state that you worked alone.]

I worked alone.

LLMs/tools and models:

[Tool names and models/versions, or explicitly state that you used none.]

Google Genimi 3.6 flash

How I used the assistance, what I learned, and how I checked it:

[Specific details for each tool and relevant problem/subpart; state not applicable if you used none.]

I used this llm to assist me in finding the proper intiger format for julia since I was unable to make it to class this week and didnt see an example notebook. I then checked it by reading the documentation for the function which I had it provide.

I also used it to debug my code in problem 1b where I accidently had a flipped sign in my constraint and forgot to add the normalization constraint. I just needed a second set of "eyes" to catch the error after I spent too long trying to find it.